METRIC

A-A-51438B
11 June 1993
SUPERSEDING
A-A-51438A
31 July 1990

COMMERCIAL ITEM DESCRIPTION

DRAINAGE UNIT, PLEURAL CAVITY

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description describes underwater and gravity chest drainage units, pleural, which accompany most heart and lung operations, and are utilized for a variety of medical conditions affecting a patient's lungs and chest. The drainage units shall be Type I, Type II or Type III as described herein.

Salient characteristics:

The drainage unit, chest, underwater or gravity, pleural, shall be a sterile, disposable, individually packaged, adult-pediatric metered thoracic drainage system. The system shall be fabricated of suitable plastic material with clearly visible chambers and diagnostic features. The Collection Chamber shall have "write on" white background columns with drainage increments correlative to Type I, II, or III units. The unit shall have holders or hooks for bed rail placement. The unit shall be wrapped in a Central Supply Room (CSR) sterile overwrap with an easy open plastic outerwrap. "Instructions For Set Up And Use" sheet shall be loose inside the plastic wrap but outside of the CSR wrap. The CSR wrap shall unwrap easily to reduce risks of contamination of the apparatus. The unit shall be adaptable to optional accessory autotransfusion recovery sets for Type I and II unit. The drainage apparatus shall include the number of chambers specified herein in each system and shall conform to the requirements of ASTM F960 for pleural space and mediasteiral drainage.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

Type I.

The container shall be divided into separate chambers as described below:

a. Suction Control Chamber: The suction control chamber shall be numerically marked in centimeters (cm) of water at 5 cm intervals to 25 (minimum), and marked from 0 to 25 cm in 1 cm intervals and shall incorporate an atmospheric vent that provides an opening for filling the suction control chamber and venting to atmosphere. The unit shall be provided with a muffler that allows air to enter the suction control chamber; a self sealing diaphragm on the face of the unit; suction regulator tubing 30 cm long (± 7.62 cm), inner diameter 0.622 cm (± 0.254 cm), outer diameter 1.092 cm (± 0.254 cm).

b. Water Seal Chamber: The water seal chamber shall act as a one way value to allow air to exit from the pleural space; serve as a manometer to measure the negativity in the patient's thoracic cavity; allow for the observation of the degree of air leak. The chamber shall have a water seal scale clearly marked for water seal level at 2 cm of water to -20 cm (minimum). A patient air leak meter shall indicate the approximate degree of air leak from the thoracic cavity. The meter shall be labeled from Low (1) to High (7). The unit shall be provided with a positive pressure relief valve to prevent pressure accumulation in excess of 2 cm; a high negativity float valve to preserve the water seal in the presence of high negativity; a filtered high negativity relief valve to vent excessive negativity.

c. Collection Chamber: The collection chamber shall have a fluid capacity of 2500 cc, and shall be calibrated in 1 cc increments up to 100 cc, and 2 cc increments up to 250 cc. Over 250 cc calibrations shall be in 5 cc increments. The surface of the chamber shall accept markings in pen or pencil. The unit shall be provided with latex collection tubing a minimum of 183 cm long, 0.927 cm inner diameter and 1.18 cm outer diameter. The collection tube shall be provided with an anti-kink coil or other design feature to prevent kinking and to ensure safe operation. The system shall allow the evacuation of air at the rate of at least 15 liters per minute.

The unit shall be provided with a floor stand that is secure, suitable and stable for routine operation, setup and use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

Type II.

Shall be mechanical suction regulator device, the unit shall have a collection chamber nominally calibrated in 5 cc increments to 500 cc, and 100 cc increments from 500 to 2500 cc maximum capacity.

The unit shall be provided with a negative pressure indicator that provides an intermittent indication of negative pressure during gravity drainage, and a constant indication of negative pressure during suction drainage.

The unit shall be provided with both positive and negative relief valves; an air leak indicator/injection port; a suction control knob with High-Low indicators; a sample port; a suction device capable of providing suction from -10 to -40 cm H_2O; the suction level shall be indicated by a float ball. The system shall allow the evacuation of air at the rate of at least 15 liters per minute.

The unit shall be provided with latex collection tubing not less than 213 cm long; inside diameter 0.927 cm, outer diameter 1.194 cm; provided with an anti-kink coil or other design feature to prevent kinking and to ensure safe operation.

Type III.

The container shall be fabricated of a suitable clear or opaque plastic material and shall be divided into separate chambers as follows:

a. Suction control chamber: The suction control chamber shall be numerically marked in centimeters of water at 5 cm intervals from 0 to at least 25 cm of water with a designated water fill line at the 20 cm mark. In order to determine the amount of suction being imposed upon the patient, dash marks between the numerical markings shall be provided at a minimum of 1cm intervals. The suction level shall be evidenced by height of fluid in the chamber.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

b. Water seal chamber: The water seal chamber shall be clearly marked for water fill level at 2 cm of water. This chamber shall allow the air to escape through the water, allowing lung expansion, while preventing the re-entry of air into the pleural cavity during periods of pleural cavity negativity as great as (-)25 cm H_2O for at least 12 seconds.

The water seal chamber, or a separate chamber devoted to patient assessment, shall serve as a manometer to measure the negativity in the patient's thoracic cavity. The manometer portion shall be numerically marked in 5 cm increments up to at least 25 cm, with dash marks at a minimum of 2 cm intervals.

c. Collection chamber, drainage: The drainage collection chamber shall have a fluid capacity of 2300 ml (±300 ml). This chamber shall be numerically marked at 10/25 ml increments, with dash marks of at least 5 ml increments for pediatric usage. A method shall be provided for an attendant to mark the chamber, indicating intermittent output for monitoring purpose.

The apparatus shall allow the evacuation of at least 16 LPM. The apparatus shall include a minimum of a 180 cm of latex rubber tubing, 9.38 mm inside diameter (ID), with a suitable plastic adapter for connection to the thoracic drainage tube. A length of tubing at least 10 cm long by at least 4.69 mm ID with a suitable plastic adapter for connection to a suction hose, source shall be provided. The apparatus shall provide positive pressure relief mechanism between water seal and suction tube to prevent buildup of pressure in excess of 2cm. If relief mechanism is part of suction hose, hose shall be marked such as "DO NOT CRIMP HOSE OR BLOCK HOSE OUTLET".

The thoracic drain tubing shall be provided with an anti-kink coil or other design feature to assure that it shall not kink at the connection to the collection chamber.

The open end(s) of the thoracic drainage tube and any other open ended tube (or open port) on the patient side of the water seal shall be provided with end protector(s)/cap(s) to prevent contamination during set-up procedure of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

Workmanship. The drainage unit, pleural shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). The unit of issue shall appear in the solicitation for Type I and Type II.

For Type III. One package containing six individually packaged sterile drainage apparatus as specified, constitutes one unit. The individual package shall be a suitable container capable of maintaining sterility of contents unless package is damaged or opened.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one package of 6 drainage units as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics. As a minimum, the following tests shall be performed on types shown.

Sampling for tests. Sampling for tests including dimensional test shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one package of 6 drainage units. Sampling shall be inspection level II with an AQL of 1.0.

For Type III. Using standard laboratory testing equipment, with the water seal chamber filled to the 2 cm level and the thoracic drain tube laying on a horizontal surface no more than 15 cm above the top of the drainage apparatus, apply a negative air pressure of (-)25 cm H_2O at the patient end of the thoracic drain tube for a 12 second time period. If air passes by the water seal, the unit is rejected.

For all Types. Perform other tests as appropriate to assure conformance with this description and with ASTM F960.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers, their Types and respective Item Identifications are covered by this document:

NATIONAL STOCK NUMBER	TYPE	ITEM IDENTIFICATION
6515-01-316-5690	I	DRAINAGE UNIT, CHEST, UNDERWATER, PLEURAL
6515-01-140-9548	II	DRAINAGE UNIT, CHEST, UNDERWATER, PLEURAL
6515-01-058-7450	III	DRAINAGE UNIT, PLEURAL CAVITY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5602
Location: c:\word\doc\nish\drain2\sg

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-51438B	11 June 1993

3. DOCUMENT TITLE
DRAINAGE UNIT, PLEURAL CAVITY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center ATTN: DPSC-MSE	(215)7372870	444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:55Z
Check the source to verify that this is the current version before use.